**Dr. Babasaheb Ambedkar Technological University, Lonere**

**B.Tech. in Chemical Engineering**

**Semester Examination: Nov 2017**

**Subject: Newer Methods of Separation (CH 704)**

**Class/Semester: Fourth Year Chemical /VII**

**Time: 3.00 Hrs** **Date: 22/11/2017** **Marks:70**

**Instructions to students**

*Question 1 is compulsory.*

*Solve **five** Questions from remaining six questions*

**Q.1** Do as directed ( 1 to 8 carries 1 M & 9 carries 2 M ) **(10 M)**

**1)** Cloud point extraction exists for....

a) Cationic surfactant b) Anionic surfactant

c) Non-ionic surfactant d) All of the above

**2)** The cloud point temperature for Tx-100 is lower or higher than Tx-114

**3)** Write down Dittus Boelter co-relation

**4)** Main disadvantage of $CO_2$ is.....

**5)** Triton X-114 full name is.....

**6)** RESS stands for.....:....

**7)** Give the range of pore size and pressure in ULTRAFILTRATION membrane process.

**8)** Give use of modifier with example.

**9)** Give the applications of SCF **(2 M)**

**Q.2. Attempt the following:**

a) Design of Cloud Point Extractor **(6 M)**

b) How GC work explain with neat diagram. **(3 M)**

c) Why calibration needs explain with example. **(3 M)**

**Q.3. Attempt the following:**

a) What types of membrane modules are commonly used? Explain the working of any one in detail with neat sketch. **(6 M)**

b) Write down the design steps for film theory. How to estimate the value of 'k' for laminar flow for tube and rectangular channel **(6 M)**

**Q.4. Attempt the following**

a) Write down the steps involve in membrane casting & casting conditions. **(4 M)**

b) Name some commonly used supercritical solvents with their critical temperature and pressure and highlights the importance of $CO_2$ among them. **(4 M)**

c) Derive thermodynamic point of view osmotic pressure equation. **(4 M)**

**Q.5. Attempt the following**

a) Write down mechanism of solubilisation of solutes from solid material. **(4 M)**

b) How DNA fragments flow through the gel ? **(4 M)**

c) Explain Demister in detail. **(4 M)**

**Q. 6. Attempt the following**

a) Draw schematic of HPLC and give its salient features. **(4 M)**

b) What is mean by Observed Retention & Real Retention. **(4 M)**

c) What are the basic forces of transport in membrane separation processes. **(4 M)**

**Q. 7. Attempt the following**

a) Derive an equation of solvent flux through reverse osmosis membrane. **(6 M)**

b) Write down the effect of temp. & effect of pH on extraction of surfactant. **(6 M)**